

static spray nozzle, and installed on a mobile platform. It is effective for deep cleaning without the need for direct human contact.

Assisting doctors and nurses.

Interestingly, Singapore has several functional cobot deployments. The country, known to be one of the most effective users of industrial robots—in the electronics sector—is now pioneering the use of cobots in spaces like healthcare. The Changi General Hospital (CGH), in association with the Centre for Healthcare Assistive and Robotics Technology (CHART), has deployed over fifty cobots in its 1000-bed hospital. The robots assist in jobs ranging from surgery and rehabilitation to delivering food and linen, lifting patients off beds and cleaning. Nurses often develop backaches a few years into the job, as they frequently lift patients. Cobots are really helpful here.

Social robots help care for the elderly patients, even playing memory games with those suffering from dementia. Some of the robots are designed to interact with people from different cultural backgrounds, in their native languages. In a media report, Selina Seah, Director of CHART, said, “One of the social robots, PARO, helped alleviate stress and anxiety so much the hospital was



Robonaut 2 onboard the ISS (Courtesy: NASA)



UR5 cobots deployed to assist employees at Aurolab (Courtesy: Universal Robots)

able to reduce its use of sedatives for dementia patients. We thought the aged patients would not take well to the robots. However, we discovered in our research that the elderly patients look at robots like life-sized toys. So, they are brought back to their childhood and, in fact, able to interact and respond better to therapy with robots than they do with a human.”

Since CGH uses robots from eight different manufacturers and running on different operating systems, it uses Robotic Middleware for Healthcare (RoMi-H), an open source system built on ROS 2, to ensure that the robots can communicate and cooperate with each other.

There have been similar deployments in some hospitals in Japan too. Last year, Hanson Robotics unveiled Grace, a humanoid robot that can assist doctors and interact with patients. In May this year, ChristianaCare, a network of private, non-profit hospitals providing healthcare services in Delaware, USA, purchased five Moxi cobots to work alongside nurses, performing nonclinical tasks such as deliveries and pickups, so that the nurses will have more free time for care delivery.

Assisting in surgeries. Bangalore based Articulus Surgical is focused on building portable and affordable surgical cobots. With appropriate tooling, collaborative robotic arms can align and hold a tool, or even provide a quick body scan, or read other vitals.

Toronto based Synaptive Medical offers a suite of robotic tools, such as point-of-care magnetic resonance imaging (MRI), automated tractography segmentation, and fully-automated robotic digital microscopy with 3D visualisation and AI-enabled voice-activated control that can help neurosurgeons.

And then there is Carlo, a cobot that can cut bones

with extreme precision using cold lasers. And, ultimately, there is the famous da Vinci surgical system. “It is a very advanced machine that can perform robot-assisted minimally-invasive surgery remotely controlled by a set of surgeons. This is a real cobot. To be frank, as I have seen in person, it is safer than a human operating, due to its sophisticated sensing, control and intelligence. The system decides and can caution the human surgeon about the best instruments, cutting depth and pressure during the surgery. Yes, cobots are perfect for healthcare and therapy. In fact, they could be very efficient and dependable assistants to doctors, paramedics and caregivers in hospital wards,” says Gurappa.

Serving food and happiness

“Cobot technology is evolving rapidly. Today’s cobots are not only used in industrial processes but also in customer-facing environments, such as in food and beverage industries where cobots are deployed to prepare and serve food at restaurants,” remarks Sougandh.

Guests are delighted when robots Champa and Chameli take orders and serve food at the Robot Chef restaurant in Bhubaneswar. Those dining at the Yellow House Robot Restaurant in Jaipur and Noida, and Chitti in Town in Hyderabad, also get to enjoy the company of robots.

While in India robot servers are deployed just to up the glee factor,